26.10.2023

Tema 02

Ex. 1:

a) ~~Copilul~~

$\Omega = \{f; b\} \times \{F; B\}$

f = copilul mic e fată

F = copilul mare e fată

b = copilul mic e băiat

B = copilul mare e băiat

a) $\cancel{\mathbb{P}(\{fF\})}$

$\cancel{\mathbb{P}(\{fF\} \mid}$

~~4 cazuri posibile: fF; fB; bF; bB.~~

$\cancel{\mathbb{P}(\{xF\}) = \frac{2}{4} = \frac{1}{2}}$ $\cancel{x \in \{f; b\}}$

$\cancel{\mathbb{P}(\{fF\}) = \frac{1}{4}}$

$\{xF\}$

2 cazuri posibile: fF; bF 1 caz favorabil: fF

$\mathbb{P}(\{fF\} \mid \{F\}) = \frac{1}{2}$

b) $fF; fB; bF$ — 3 cazuri posibile

fF — 1 caz favorabil

$\mathbb{P}(\{fF\} \mid$ cel puțin unul din copii e fată$) = \frac{1}{3}$

<u>Ex. 2:</u>

Probabilitatea teoretică:

Nr. cazuri posibile $2^n = \left|\{\text{cap; pajură}\}^n\right|$

Nr. cazuri favorabile: exact k monezi cap $\Rightarrow$

$\Rightarrow n-k$ monezi sunt pajură $\Rightarrow$

$\Rightarrow C_n^k$

$\mathbb{P} = \frac{C_n^k}{2^n}$; pt. $k \leq n$, $k, n \in \mathbb{N}$

$\mathbb{P} = \frac{\frac{n!}{k!(n-k)!}}{2^n} = \frac{n!}{k!(n-k)!\,2^n}$

<u>Ex. 4:</u>

B = bolnav

t_1 = primul test pozitiv

t_2 = al doilea test pozitiv

a) $\mathbb{P}(B \mid t_1 \cap t_2^c) = ?$

b) $\mathbb{P}(B^c \mid t_1^c \cap t_2^c) = ?$

a) $\mathbb{P}(B \mid t_1 \cap t_2^c) = \frac{\mathbb{P}(B \cap t_1 \cap t_2^c)}{\mathbb{P}(t_1 \cap t_2^c)} =$

$= \frac{\mathbb{P}(B) \cdot \mathbb{P}(t_1 \cap t_2^c / B)}{\mathbb{P}(t_1 \cap t_2^c \cap B) + \mathbb{P}(t_1 \cap t_2^c \cap B^c)} =$

$$= \frac{\mathbb{P}(B) \cdot \mathbb{P}(t_1 \cap t_2^c \mid B)}{\mathbb{P}(B) \cdot \mathbb{P}(t_1 \cap t_2^c \mid B) + \mathbb{P}(B^c) \cdot \mathbb{P}(t_1 \cap t_2^c \mid B^c)}$$

$$= \frac{\mathbb{P}(B) \cdot \mathbb{P}(t_1 \mid B) \cdot \mathbb{P}(t_2^c \mid B)}{\mathbb{P}(B) \cdot \mathbb{P}(t_1 \mid B) \cdot \mathbb{P}(t_2^c \mid B) + \mathbb{P}(B^c)\, \mathbb{P}(t_1 \mid B^c)\, \mathbb{P}(t_2^c \mid B^c)}$$

$$= \frac{\frac{1}{1000} \cdot \frac{98}{100} \cdot \frac{2}{100}}{\frac{1}{1000} \cdot \frac{98}{100} \cdot \frac{2}{100} + \frac{999}{1000} \cdot \frac{5}{100} \cdot \frac{95}{100}} = \frac{98 \cdot 2}{98 \cdot 2 + 999 \cdot 5 \cdot 95} \simeq$$

$$\simeq \frac{196}{196 + 474525} = \frac{196}{474721} \simeq$$

$$\simeq 0{,}000411 = 0{,}041\%$$

$$\mathbb{P}(B) = \frac{1}{1000} \Rightarrow \mathbb{P}(B^c) = \frac{999}{1000}$$

$$\mathbb{P}(t_1 \mid B) = \frac{98}{100} = \mathbb{P}(t_2 \mid B)$$

$$\mathbb{P}(t_1 \mid B^c) = \frac{5}{100} = \mathbb{P}(t_2 \mid B^c)$$

$$\cancel{\mathbb{P}(t_1 \mid B^c)}$$

$$\mathbb{P}(t_2^c \mid B) = 1 - \mathbb{P}(t_2 \mid B) = \frac{2}{100}$$

$$\mathbb{P}(t_2^c \mid B^c) = 1 - \mathbb{P}(t_2 \mid B^c) = \frac{95}{100}$$

$$\cancel{\mathbb{P}(t_2^c}$$

$$\mathbb{P}(t_1^c \mid B^c) = \frac{95}{100}$$

$$\mathbb{P}(t_1^c \mid B) = \frac{2}{100}$$

b)

$$\mathbb{P}(B^c \mid +_1^c \cap +_2^c) = \frac{\mathbb{P}(B^c \cap +_1^c \cap +_2^c)}{\mathbb{P}(+_1^c \cap +_2^c)} =$$

$$= \frac{\mathbb{P}(B^c) \cdot \mathbb{P}(+_1^c \cap +_2^c \mid B^c)}{\mathbb{P}(+_1^c \cap +_2^c \cap B) + \mathbb{P}(+_1^c \cap +_2^c \cap B^c)} =$$

$$= \frac{\mathbb{P}(B^c) \cdot \mathbb{P}(+_1^c \cap +_2^c \mid B^c)}{\mathbb{P}(B^c) \cdot \mathbb{P}(+_1^c \cap +_2^c \mid B^c) + \mathbb{P}(B) \cdot \mathbb{P}(+_1^c \cap +_2^c \mid B)} =$$

$$= \frac{\mathbb{P}(B^c)\,\mathbb{P}(+_1^c \mid B^c)\,\mathbb{P}(+_2^c \mid B^c)}{\mathbb{P}(B^c)\,\mathbb{P}(+_1^c \mid B^c) \cdot \mathbb{P}(+_2^c \mid B^c) + \mathbb{P}(B)\,\mathbb{P}(+_1^c \mid B)\,\mathbb{P}(+_2^c \mid B)} =$$

$$= \frac{\frac{999}{1000} \cdot \frac{95}{100} \cdot \frac{95}{100}}{\frac{999}{1000} \cdot \frac{95}{100} \cdot \frac{95}{100} + \frac{1}{1000} \cdot \frac{2}{100} \cdot \frac{2}{100}} =$$

$$\cancel{\mathbb{P}(B^c) = \frac{999}{1000}}$$

$$= \frac{999 \cdot 95 \cdot 95}{999 \cdot 95 \cdot 95 + 4} =$$

$$= \frac{9015975}{9015975 + 4} = \cancel{\frac{9015975}{9016015} \approx 0{,}999996} =$$

$$= \cancel{99{,}9996\%}$$

$$= \frac{9015975}{9015979} \approx 0{,}999996 = 99{,}9996\%$$